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Thalamotomy

Number: 0153

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Scope of Policy

This Clinical Policy Bulletin addresses thalamotomy.

I. **Medical Necessity**

A. Aetna considers unilateral thalamotomy medically necessary for abolishing tremor and rigidity in members with movement disorders, including dystonia, Parkinson's disease, spasmodic torticollis, tremor; and who meet *all* of the following selection criteria:

1. Member has a history of positive response to drug therapy (i.e., member had a positive initial response to medication, but has subsequently become refractory); *and*
2. Member has been screened by a neurologist who has expertise in movement disorders to ensure all appropriate non-surgical therapies have been tried; *and*
3. Member has severe and incapacitating tremors and medical therapy has failed as indicated by worsening of symptoms and/or disabling medication side effects.

B. Aetna considers focused ultrasound thalamotomy medically necessary for severe essential tremor (ET) inadequately responsive to medical therapy

C. Aetna considers gamma knife thalamotomy medically necessary for the treatment of the following indications:

1. Severe essential tremor (ET) inadequately responsive to medical therapy; *or*
2. Refractory disabling tremor and rigidity from Parkinson's disease in persons who meet medical necessity criteria in section above.
3. Last resort for the treatment of malignant pain when *all* of the following selection criteria are met:

a. Advanced oncological disease with limited life expectancy; *and*
b. Options for radiotherapy have been exhausted; *and*
c. Best medical treatment has failed to cause pain relief or incurred the development of intolerable side effects; *and*

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d. Pain has failed to respond adequately to any targeted interventions (e.g., nerve blocks); *and*
e. Absence of technical limitations or medical contraindications to thalamotomy.

II. Experimental, Investigational, or Unproven

The following procedures are considered experimental, investigational, or unproven because the effectiveness of these approaches has not been established:

- Bilateral thalamotomy - according to guidelines from the American Academy of Neurology, bilateral (second side) thalamotomy is not recommended because of adverse side effects;
- Focused ultrasound thalamotomy - for the treatment of fragile X-associated tremor/ataxia syndrome, multiple sclerosis-associated tremor, non-ET tremor syndromes (e.g., dystonic tremor, dystonia gene-associated tremor, and Parkinson's disease), and obsessive-compulsive disorder;
- Magnetic resonance-guided focused ultrasound thalamotomy for the treatment of refractory neuropathic pain;
- Sub-thalamic nucleus focused ultrasound thalamotomy - for the treatment of post-operative dystonia;
- Thalamotomy - for the treatment of non-malignant pain or other indications;
- Ventro-oral thalamotomy - for the treatment of focal hand dystonia;
- Ventro-oral focused ultrasound thalamotomy - for the treatment of segmental dystonia (including cervical dystonia).

III. Related Policies

- CPB 0208 - Deep Brain Stimulation - for deep brain stimulation for the treatment of essential tremor and Parkinson's disease
- CPB 0307 - Parkinson's Disease - for pallidotomy for the treatment of Parkinson's disease

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Table:

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
***Thalamotomy*:**	
CPT codes covered if selection criteria are met:	
61720	Creation of lesion by stereotactic method, including burr hole(s) and localizing and recording techniques, single or multiple stages; globus pallidus or thalamus
ICD-10 codes covered if selection criteria are met:	
G20.A1 - G20.C	Parkinson's disease
G21.0 - G21.9	Secondary Parkinsonism
G24.01 - G24.9	Dystonia
G24.3	Spasmodic torticollis

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Code	Code Description
G25.0 - G25.9	Other extrapyramidal and movement disorders
G89.3	Neoplasm related pain (acute) (chronic)

ICD-10 codes not covered for indications listed in the CPB (not all-inclusive):

F42.2 - F42.9	Obsessive-compulsive disorder
G89.0 - G89.29	Pain, not elsewhere classified
G89.4	Chronic pain syndrome
R52	Pain, unspecified

Gamma knife thalamotomy:

CPT codes covered if selection criteria are met:

61796	Stereotactic radiosurgery (particle beam, gamma ray or linear accelerator), 1 simple cranial lesion
+ 61797	each additional cranial lesion, simple (List separately in addition to code for primary procedure)
61798	1 complex cranial lesion
+ 61799	each additional cranial lesion, complex (List separately in addition to code for primary procedure)

ICD-10 codes covered if selection criteria are met:

G20.A1 - G21.9	Parkinson's disease
G25.0 - G25.9	Other extrapyramidal and movement disorders

Focused ultrasound thalamotomy:

CPT codes covered if selection criteria are met:

61715	Magnetic resonance image guided high intensity focused ultrasound (MRgFUS), stereotactic ablation of target, intracranial, including stereotactic navigation and frame placement, when performed

CPT codes not covered if selection criteria are met:

***Ventro-oral focused ultrasound thalamotomy and sub-thalamic nucleus focused ultrasound thalamotomy*-no specific code**

HCPCS codes not covered if selection criteria are met:

C9734	Focused ultrasound ablation/therapeutic intervention, other than uterine leiomyomata, with magnetic resonance (MR) guidance

ICD-10 codes covered if selection criteria are met:

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Code	Code Description
G25.0	Essential tremor

ICD-10 codes not covered if selection criteria are met:

G20.A1 - G21.C	Parkinson's disease
G21.0 - G21.9	Secondary Parkinsonism
G24.01 - G24.9	Dystonia
G25.1	Drug-induced tremor [multiple sclerosis-associated tremor]
G25.2	Other specified forms of tremor [multiple sclerosis-associated tremor]
G56.00 - G56.93	Mononeuropathies of upper limb
G57.00 - G57.93	Mononeuropathies of lower limb
G58.0 - G58.9	Other mononeuropathies
M54.10 - M54.18	Radiculopathy
M54.40 - M54.42	Lumbago with sciatica
M79.2	Neuralgia and neuritis, unspecified
Q99.2	Fragile X syndrome

Ventro-oral thalamotomy:

CPT codes not covered for indications listed in the CPB:

Ventro-oral thalamotomy - no specific code:

ICD-10 codes not covered for indications listed in the CPB:

G24.8	Other dystonia [focal hand dystonia]

Background

Thalamotomy, a surgical intervention for the treatment of various forms of movement disorders such as Parkinson's disease, tremor, and dystonia, is a procedure that severs nerve fibers from an area of the brain called the thalamus.

Movement disorders are often caused by lesions of the extrapyramidal motor system. These disorders are characterized by involuntary movements including tremor, dystonia, chorea, athetosis, and hemiballism. In the early 1960s, stereotactic thalamotomy with the ventrolateral nucleus as the target site was the treatment of choice for a wide variety of movement disorders, including the tremor and rigidity associated with Parkinson’s disease (PD). The discovery of levodopa in the late 1960s, however, prompted the preferential use of pharmacotherapies over neuroablative procedures in the treatment of movement disorders in the last 35 years. The limitations of drug therapy as well as the improvements in imaging (computerized tomography and magnetic resonance imaging) and electrophysiological recording techniques (microelectrode-guided mapping) led to renewed interests in the use of stereotactic thalamotomy for the management of patients with

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movement disorders. Available evidence indicates that thalamotomy is an effective procedure in treating patients with movement disorders, especially for individuals with essential tremor or those with tremor secondary to PD.

Thalamotomy is effective in treating tremor, but has little or no effect on akinesia or bradykinesia. For PD patients with symptoms other than tremor, pallidotomy is preferred over thalamotomy. It is not surprising that both thalamotomy and pallidotomy have similar effects on tremor since the thalamic ventral nuclear group receives efferent projection from the globus pallidus.

Stereotactic thalamotomy is always carried out under local anesthesia. The target site is delineated by means of a computed tomography (CT) or magnetic resonance imaging (MRI) scan performed with a stereotactic frame attached to the head. In general, all types of tremor are best treated by lesions located in the ventralis intermedius nucleus (part of the ventralis lateralis nucleus) just anterior to the sensory relay nucleus. Hypertonic disorders such as hemiballismus supposedly will respond to more anteriorly located lesions, in the anterior ventralis oralis posterior nucleus (part of the ventralis lateralis nucleus) or the ventralis oralis anterior nucleus. The coordinates of the target site are determined in reference to a line drawn between the anterior commissure-posterior commissure (AC-PC line). The typical coordinates for the ventralis intermedius nucleus (for the treatment of tremor) are usually 4 mm behind the midpoint of the AC-PC line, 13 mm lateral to midline, and 1 mm above the AC-PC line. Proportional adjustments are made in relation to the length of the AC-PC line of the patient. When the target coordinates have been defined, the patient is mildly sedated. Under local anesthetic, the target site is reached through a frontal burr hole placed 1 cm anterior to the coronal suture and 3 cm lateral to the sagittal suture. An insulated stimulating electrode is then inserted under impedance monitoring into the ventralis intermedius nucleus. The target zone is stimulated with small electrical impulses, the goal of which is to ensure that the probe is in the correct location of the thalamus. With electrical stimulation, tremor and rigidity can be reduced almost immediately and this confirms accurate placement of the probe. Electrostimulation may cause untoward symptoms indicating that the electrode tip may need repositioning.

Young and colleagues (2000) investigated the long-term effects of gamma knife thalamotomy (GKT) for treatment of disabling tremor. A total of 158 patients underwent MRI-guided radiosurgical nucleus ventralis intermedius (VIM) thalamotomy for the treatment of parkinsonian tremor (n = 102), essential tremor (n = 52), or tremor due to stroke, encephalitis, or cerebral trauma (n = 4). Pre-operative and post-operative blinded assessments were performed by a team of independent examiners skilled in the evolution of movement disorders. A single isocenter exposure with the 4-mm collimator helmet of the Leksell gamma knife unit was used to make the lesions. In patients with Parkinson's disease 88.3 % became fully or nearly tremor-free, with a mean follow up of 52.5 months. Statistically significant improvements were seen in Unified Parkinson's Disease Rating Scale tremor scores and rigidity scores, and these improvements were maintained in 74 patients followed 4 years or longer. In patients with essential tremor, 92.1 % were fully or nearly tremor-free post-operatively, but only 88.2 % remained tremor-free by 4 years or more post-GKT. Statistically significant improvements were seen in the Clinical Rating Scale for tremor in essential tremor patients and these improvements were well maintained in the 17 patients, followed 4 years or longer. Only 50 % of patients with tremor of other origins improved significantly. One patient sustained a transient complication and 2 patients sustained mild permanent side effects from the treatments. The authors concluded that GKT (at the VIM) provided relief from tremor equivalent to that provided by radiofrequency thalamotomy or deep brain stimulation, but it is safer than either of these alternatives. Additionally, long-term follow-up indicated that relief of tremor is well maintained. No long-term radiation-induced complications have been observed.

Niranjan and associates (2000) reported their findings of 12 patients (median age of 75 years) who underwent GKT for essential tremor (n = 9) or multiple sclerosis-related tremor (n = 3). All 11 evaluable patients noted improvement in action tremor; 6 of 8 essential tremor patients had complete tremor arrest, and the violent action tremor in all 3 patients with multiple sclerosis was improved. One patient developed transient arm weakness. Stereotactic radiosurgery for essential tremor and multiple sclerosis-related tremor is safe and effective for patients who may be poor candidates for other procedures. The findings by Young et al (2000) and Niranjan et al (2000) were in agreement with that of Ohye et al (2000) who reported that GKT appeared to be an alternative useful method in selected cases of parkinsonian and other tremors (n = 36).

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Ohye and co-workers (2005) studied the effects of GKT on PD-related tremor and essential tremor before and after reloading of radioactive cobalt. Based on experience in stereotactic thalamotomy aided by depth microrecording, the target was located at the lateral border of the thalamic VIM. For more precise targeting, the percentage representation of the thalamic VIM in relation to the entire thalamic length is useful. The location of the target was determined on MRI and computerized tomography scanning. A maximum dose of 130 Gy was delivered to the target by using a single isocenter with the 4-mm collimator. In more recent cases, a systematic follow-up examination was performed at 3, 6, 12, 18, and 24 months after GKT. Since 1993, the authors have treated 70 patients with PD. Throughout the series the same dosimetric technique has been used. The course after GKT was compared between the 25 cases with PD treated before reloading and the 35 cases treated after reloading. In the majority (80 to 85 %) treated after reloading, tremor and rigidity were reduced around 6 months after GKT. In the cases treated before reloading this effect took approximately 1 year.

Mathieu et al (2007) discussed their experience with GKT in the management of 6 consecutive patients suffering from disabling multiple sclerosis tremor. The median age at the time of radiosurgery was 46 years (range of 31 to 57 years). Intention tremor had been present for a median of 3 years (range of 8 months to 12 years). One 4-mm isocenter was used to deliver a median maximum dose of 140 Gy (range of 130 to 150 Gy) to the VIM of the thalamus opposite the side of the most disabling tremor. Clinical outcome was assessed using the Fahn-Tolosa-Marin scale. The median follow-up was 27.5 months (range of 5 to 46 months). All patients experienced improvement in tremor after a median latency period of 2.5 months. More improvement was noted in tremor amplitude than in writing and drawing ability. In 4 patients, the tremor reduction led to functional improvement. One patient suffered from transient contralateral hemiparesis, which resolved after brief corticosteroid administration. No other complication was seen. The authors concluded that GKT is effective as a minimally invasive alternative to stereotactic surgery for the palliative treatment of disabling multiple sclerosis tremor.

Kondziolka et al (2008) evaluated the results following GKT for medically refractory essential tremor in a series of patients in whom open surgical techniques were not desirable. A total of 31 patients underwent GKT for disabling essential tremor after medical therapy had failed. Their mean age was 77 years. Most patients were elderly or had concomitant medical illnesses. A single 4-mm isocenter was used to target a maximum dose of 130 or 140 Gy to the VIM. Items from the Fahn-Tolosa-Marin clinical tremor rating scale were used to grade tremor and handwriting before and after radiosurgery. The median follow-up was 36 months. In the group of 26 evaluable patients, the mean tremor score (+/- standard deviation) was 3.7 +/- 0.1 pre-operatively and 1.7 +/- 0.3 after radiosurgery ($p < 0.000015$). The mean handwriting score was 2.8 +/- 0.2 before GKT and 1.7 +/- 0.2 afterward ($p < 0.0002$). After radiosurgery, 18 patients (69 %) showed improvement in both action tremor and writing scores, 6 (23 %) only in action tremor scores, and 3 (12 %) in neither tremor nor writing. Permanent mild right hemiparesis and speech impairment developed in 1 patient 6 months after radiosurgery. Another patient had transient mild right hemiparesis and dysphagia. The authors concluded that GKT is a safe and effective therapy for medically refractory essential tremor. Its use is especially valuable for patients ineligible for radiofrequency thalamotomy or deep brain stimulation. Patients must be counseled on potential complications, including the low probability of a delayed neurological deficit.

Duma (2007) stated that GKT is an effective and useful alternative to invasive radiofrequency techniques for patients with movement disorder who are at high surgical risk. The mechanical accuracy of the gamma unit combined with the anatomical accuracy of high-resolution MRI make radiosurgical lesioning safe and precise. Furthermore, higher radiosurgical doses are more effective than lower ones at eliminating or reducing tremor, and are generally without complications.

Friehs et al (2007) noted that stereotactic radiosurgery (SRS) with the gamma knife and linear accelerator has revolutionized neurosurgery over the past 20 years. The most common indications for radiosurgery today are tumors and arteriovenous malformations of the brain. Functional indications such as treatment of movement disorders or intractable pain only contribute a small percentage of treated patients. The authors stated that radiosurgical ventrolateral thalamotomy for the treatment of tremor in patients with PD or multiple sclerosis, as well as in the treatment of essential tremor, may be indicated for a select group of patients with advanced

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age, significant medical conditions that preclude treatment with open surgery, or patients who must receive anti-coagulation therapy.

Critical outcome measures deemed important in assessing the effectiveness of thalamotomy in the treatment of patients with movements disorders are reduction or disappearance of tremor and rigidity, improvement in motor function, and/or reduction in the consumption of anti-parkinsonian or tremor medications.

There is sufficient scientific evidence that thalamotomy can alleviate or abolish tremor and rigidity in properly selected patients with movement disorders including PD, dystonia, tremor, and multiple sclerosis.

Appropriate candidates for thalamotomy are patients with severe and incapacitating tremor who have tried and failed medical therapy as indicated by worsening of symptoms and/or disabling medication side effects. Patients should have a history of positive response to drug therapy (i.e., positive initial response, then became refractory to medication). Patients should be screened by a neurologist who has expertise in movement disorders to ensure all reasonable forms of pharmacotherapies have been tried and failed.

In a review on destructive procedures for the treatment of non-malignant pain, Cetas and colleagues (2008) reviewed the following ablative procedures: cingulotomy, cordotomy, dorsal root entry zone (DREZ) lesioning, ganglionectomy, mesencephalotomy, myelotomy, neurotomy, rhizotomy, sympathectomy, thalamotomy, and tractotomy. Articles related to pain resulting from malignancy and those not in peer-reviewed journals were excluded. In reviewing pertinent articles, focus was placed on patient number, outcome, and follow-up. A total of 146 articles was included in the review. The majority of studies (n = 131) constituted Class III evidence. Eleven Class I and 4 Class II studies were found, of which nearly all (13 of 15) evaluated radiofrequency rhizotomies for different pain origins, including lumbar facet syndrome, cervical facet pain, and Type I or typical trigeminal neuralgia. Overall, support for ablative procedures for non-malignant pain is derived almost entirely from Class III evidence; despite a long history of use in neurosurgery, the evidence supporting destructive procedures for benign pain conditions remains limited. The authors concluded that newly designed prospective standardized studies are needed to define indications and outcomes for these procedures.

According to available literature, thalamotomy is contraindicated in *any* of the following circumstances where the safety and effectiveness of thalamotomy have not been established:

- Persons with dementia or cerebral atrophy; *or*
- Persons with Parkinson's plus or atypical Parkinson's disorders (e.g. multi-system atrophy involving the striatum, cerebellum, pons, and medulla such as striatonigral degeneration, olivoponto-cerebellar degeneration, progressive supranuclear palsy, or combined Alzheimer's and Parkinson's disease); *or*
- Persons with very advanced PD (Hoehn and Yahr stage V).

An UpToDate review on “Surgical treatment of essential tremor” (Tarsy, 2016) states that "the long-term effectiveness and safety of this procedure [ultrasound thalamotomy] remain uncertain and warrant further study.”

In a proof-of-concept study, Lipsman et al (2013) examined MRI-guided focused ultrasound thalamotomy to the management of essential tremor (ET). This study was done in Toronto, Canada, between May, 2012, and January, 2013. A total of 4 patients with chronic and medication-resistant ET were treated with MRI-guided focused ultrasound to ablate tremor-mediating areas of the thalamus. Patients underwent tremor evaluation and neuroimaging at baseline and 1 month and 3 months after surgery. Outcome measures included tremor severity in the treated arm, as measured by the clinical rating scale for tremor, and treatment-related adverse events. Patients showed immediate and sustained improvements in tremor in the dominant hand. Mean reduction in tremor score of the treated hand was 89.4 % at 1 month and 81.3 % at 3 months. This reduction was accompanied by functional benefits and improvements in writing and motor tasks. One patient had post-operative paraesthesias, which persisted at 3 months. Another patient developed a deep vein thrombosis, potentially related to the length of the procedure. The authors concluded that MRI-guided focused ultrasound might be a safe and effective approach to generation of focal intracranial lesions for the management of disabling, medication-resistant ET. They stated that if larger trials validate the safety and

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ascertain the effectiveness and durability of this new approach, it might change the way that patients with ET and potentially other disorders are treated.

In an open-label, pilot study, Elias et al (2013) examined the use of transcranial MRI-guided focused ultrasound thalamotomy for the treatment of ET. From February 2011 through December 2011, these investigators used transcranial MRI-guided focused ultrasound to target the unilateral ventral intermediate nucleus of the thalamus in 15 patients with severe, medication-refractory ET. They recorded all safety data and measured the effectiveness of tremor suppression using the Clinical Rating Scale for Tremor to calculate the total score (ranging from 0 to 160), hand sub-score (primary outcome, ranging from 0 to 32), and disability sub-score (ranging from 0 to 32), with higher scores indicating worse tremor. These researchers assessed the patients' perceptions of treatment effectiveness with the Quality of Life in Essential Tremor Questionnaire (ranging from 0 to 100 %, with higher scores indicating greater perceived disability). Thermal ablation of the thalamic target occurred in all patients. Adverse effects of the procedure included transient sensory, cerebellar, motor, and speech abnormalities, with persistent paresthesias in 4 patients. Scores for hand tremor improved from 20.4 at baseline to 5.2 at 12 months ($p = 0.001$). Total tremor scores improved from 54.9 to 24.3 ($p = 0.001$). Disability scores improved from 18.2 to 2.8 ($p = 0.001$). Quality-of-life scores improved from 37 % to 11 % ($p = 0.001$). The authors concluded that in this pilot study, ET improved in 15 patients treated with MRI-guided focused ultrasound thalamotomy. Moreover, they stated that large, randomized, controlled trials are needed to evaluate the procedure's safety and effectiveness. The drawbacks of this study included:

I. lack of a control group,
II. comprehensive cognitive assessments were not performed; and it is possible that focused ultrasound thalamotomy resulted in cognitive impairment, and
III. patients and researchers were all aware of treatments that were performed, which may have introduced bias in favor of reporting improvements in symptoms and quality of life.

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II. comprehensive cognitive assessments were not performed; and it is possible that focused ultrasound thalamotomy resulted in cognitive impairment, and
III. patients and researchers were all aware of treatments that were performed, which may have introduced bias in favor of reporting improvements in symptoms and quality of life.

Chang et al (2015) noted that recently magnetic resonance-guided focused ultrasound (MRgFUS) has been developed as a less-invasive surgical tool aimed to precisely generate focal thermal lesions in the brain. In this feasibility study, patients underwent tremor evaluation and neuroimaging study at baseline and up to 6 months after MRgFUS. Tremor severity and functional impairment were assessed at baseline and then at 1 week, 1 month, 3 months, and 6 months after treatment. Adverse effects were also sought and ascertained by directed questions, neuroimaging results and neurological examination. The current feasibility study attempted MRgFUS thalamotomy in 11 patients with medication-resistant ET. Among them, 8 patients completed treatment with MRgFUS, whereas 3 patients could not complete the treatment because of insufficient temperature. All patients who completed treatment with MRgFUS showed immediate and sustained improvements in tremors lasting for the 6-month follow-up period. Skull volume and maximum temperature rise were linearly correlated (linear regression, $p = 0.003$). Other than 1 patient who had mild and delayed post-operative balance, no patient developed significant post-surgical complications; about 50 % of the patients had bouts of dizziness during the MRgFUS. The authors concluded that these results demonstrated that MRgFUS thalamotomy is a safe, effective and less-invasive surgical method for treating medication-refractory ET. However, they stated that several issues must be resolved before clinical application of MRgFUS, including optimal patient selection and management of patients during treatment.

Jung et al (2015) reported different MRI patterns in patients with essential tremor (ET) or obsessive-compulsive disorder (OCD) after transcranial MRgFUS and discussed possible causes of occasional MRgFUS failure. Between March 2012 and August 2013, MRgFUS was used to perform unilateral thalamotomy in 11 ET patients and bilateral anterior limb capsulotomy in 6 OCD patients; in all patients symptoms were refractory to drug therapy. Sequential MR images were obtained in patients across a 6-month follow-up period. For OCD patients, lesion size slowly increased and peaked 1 week after treatment,

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after which lesion size gradually decreased. For ET patients, lesions were visible immediately after treatment and markedly reduced in size as time passed. In 3 ET patients and 1 OCD patient, there was no or little temperature rise (i.e., less than 52°C) during MRgFUS. Successful and failed patient groups showed differences in their ratio of cortical-to-bone marrow thickness (i.e., skull density). The authors found different MRI pattern evolution after MRgFUS for white matter and gray matter. These results suggested that skull characteristics, such as low skull density, should be evaluated prior to MRgFUS to successfully achieve thermal rise.

There is currently insufficient evidence to support the use of MRgFUS for the treatment of ET and OCD.

Schlesinger and co-workers (2015) investigated the effectiveness of MRgFUS for moderate-to-severe tremor in PD. A total of 7 patients (mean age of 59.4 ± 9.8 years, range of 46 to 74) with a mean disease duration of 5.4 ± 2.8 years (range of 2 to 10) suffering from severe refractory tremor, underwent VIM thalamotomy using MRgFUS. Tremor stopped in the contralateral upper extremity in all patients immediately following treatment. Total Unified Parkinson's Disease Rating Scale (UPDRS) decreased from 37.4 ± 12.2 to 18.8 ± 11.1 ($p = 0.007$) and PDQ-39 decreased from 42.3 ± 16.4 to 21.6 ± 10.8 ($p = 0.008$) following MRgFUS. These effects were sustained (mean follow-up of 7.3 months). Adverse events (AEs) during MRgFUS included headache (n = 3), dizziness (n = 2), vertigo (n = 4), and lip paresthesia (n = 1) and following MRgFUS were hypogeusia (n = 1), unsteady feeling when walking (n = 1, resolved), and disturbance when walking tandem (n = 1, resolved). The authors concluded that thalamotomy using MRgFUS is safe and effective in PD patients; however, large randomized studies are needed to evaluate prolonged safety and effectiveness.

Elias and colleagues (2016) noted that uncontrolled pilot studies have suggested the effectiveness of MRgFUS for the treatment of ET. These investigators enrolled patients with moderate-to-severe ET that had not responded to at least 2 trials of medical therapy and randomly assigned them in a 3:1 ratio to undergo unilateral focused ultrasound thalamotomy or a sham procedure. The Clinical Rating Scale for Tremor (CRST) and the Quality of Life in Essential Tremor Questionnaire (QUEST) were administered at baseline and at 1, 3, 6, and 12 months. Tremor assessments were videotaped and rated by an independent group of neurologists who were unaware of the treatment assignments. The primary outcome was the between-group difference in the change from baseline to 3 months in hand tremor, rated on a 32-point scale (with higher scores indicating more severe tremor). After 3 months, patients in the sham-procedure group could cross-over to active treatment (the open-label extension cohort). A total of 76 patients were included in the analysis. Hand-tremor scores improved more after focused ultrasound thalamotomy (from 18.1 points at baseline to 9.6 at 3 months) than after the sham procedure (from 16.0 to 15.8 points); the between-group difference in the mean change was 8.3 points (95 % confidence interval [CI]: 5.9 to 10.7; $p < 0.001$). The improvement in the thalamotomy group was maintained at 12 months (change from baseline, 7.2 points; 95 % CI: 6.1 to 8.3). Secondary outcome measures assessing disability and quality of life also improved with active treatment (the blinded thalamotomy cohort) as compared with the sham procedure ($p < 0.001$ for both comparisons); AEs in the thalamotomy group included gait disturbance in 36 % of patients and paresthesias or numbness in 38 %; these AEs persisted at 12 months in 9 % and 14 % of patients, respectively. The authors concluded that MRgFUS reduced hand tremor in patients with ET; side effects included sensory and gait disturbances.

This study had several drawbacks:

I. the procedures were all performed unilaterally. Although unilateral focused ultrasound thalamotomy improved total tremor scores by 47 % in the study cohort, there was no reduction of ipsilateral tremor and only minimal improvement in axial tremors of the head, neck, and voice,
II. some patients may be reluctant or unwilling to undergo MRI studies or it may be unsafe for them to do so,
III. lesioning procedures require a balance between the size of the lesion and the risk of AEs, since larger lesions are expected to have more enduring efficacy but a greater incidence of side effects, and
IV. transcranial delivery of focused ultrasound was difficult to achieve in 5 of the study patients, probably because of the frequency and other properties of the acoustic wave, as well as individual cranial characteristics; additional research is needed to address this issue.

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IV. transcranial delivery of focused ultrasound was difficult to achieve in 5 of the study patients, probably because of the frequency and other properties of the acoustic wave, as well as individual cranial characteristics; additional research is needed to address this issue.

Moreover, the authors stated that the benefits and risks of focused ultrasound thalamotomy performed in a carefully controlled clinical trial may differ from the benefits and risks with routine practice in diverse clinical settings.

An accompanying editorial (Louis, 2016) noted additional study limitations. The first is the limited follow-up period, which was 12 months. The editorialist stated that sustained benefit at 2 years, 3 years, and 5 or more years is not known. Studies with longer follow-up intervals are needed to address this issue. This is particularly important because of tachyphylaxis, which is the second concern. The primary outcome measure was the score for hand tremor (on a scale ranging from 0 to 32, with higher scores indicating more severe tremor) at 3 months. The editorialist noted that the tremor score in the group that underwent focused ultrasound thalamotomy increased from 8.84 (at 1 month) to 9.55 (at 3 months) to 10.13 (at 6 months) to 10.89 (at 12 months). The increase from months 1 to 12 was 23%. The editorialist noted that secondary outcome measures showed similar or greater increases during the 12-month follow-up period (e.g., the Clinical Rating Scale for Tremor score increased from 23.38 at 1 month to 32.38 at 12 months, which is a 38% increase). The editorialist stated that it was not clear whether this loss of efficacy, which is also seen to some extent with deep-brain stimulation, is due to disease progression or tolerance, although typical estimates of the rate of disease progression in essential tremor make the former possibility less likely.

In addition, adverse events at three months were more common in the thalamotomy group, including gait disturbance in 36 percent and numbness or paresthesia in 38 percent; these persisted at 12 months in 9 and 14 percent, respectively (Okun, 2016). Ultrasound thalamotomy produces a nonreversible brain lesion and should not be performed bilaterally due to associated speech and swallowing effects.

In a double-blinded, randomized controlled trial (RCT), Bond and associates (2016) examined the effectiveness of MRgFUS thalamotomy in tremor-dominant PD. Patients with medication-refractory, tremor-dominant PD were enrolled in the 2-center study and randomly assigned 1:2 to receive either a sham procedure or treatment. After the 3-month blinded phase, the sham group was offered treatment. Outcome was measured with blinded CRST and UPDRS ratings. The primary outcome compared improvement in hand tremor between the treatment and sham procedure at 3 months. Secondary outcomes were measured with UPDRS and hand tremor at 12 months. Safety was assessed with MRI, AEs, and comprehensive neurocognitive assessment. A total of 27 patients were enrolled and 6 were randomly assigned to a sham procedure. For the primary outcome assessment, there was a mean 50 % improvement in hand tremor from MRgFUS thalamotomy at 3 months compared with a 22 % improvement from the sham procedures ($p = 0.088$). The 1-year tremor scores for all 19 patients treated with 1-year follow-up data (blinded and un-blinded) showed a reduction in tremor scores of 40.6 % ($p = 0.0154$) and a mean reduction in medicated UPDRS motor scores of 3.7 (32 %, $p = 0.033$). Sham patients had a notable placebo effect with a mean 21.5 % improvement in tremor scores at 3 months; 27 patients completed the primary analysis, 19 patients completed the 12-month assessment, 3 patients opted for deep brain stimulation (DBS), 3 were lost to follow-up, 1 patient opted for no treatment, and 1 is pending a 12-month evaluation. The authors concluded that transcranial MRgFUS demonstrated a trend toward improvement in hand tremor, and a clinically significant reduction in mean UPDRS. They stated that a significant placebo response was noted in the randomized trial.

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MRI-Guided Focused Ultrasound Thalamotomy in Fragile X-Associated Tremor / Ataxia Syndrome

Fasano and colleagues (2016) stated that MRgFUS is a promising, incision-free but nevertheless invasive technique to ablate deep brain targets. Recent studies have examined the safety and effectiveness of MRgFUS targeting the VIM of patients with tremor. In separate studies, 4, 15, and 8 patients with ET were included and followed-up for 3 to 12 months after unilateral MRgFUS. The majority of the cases had a clinically meaningful reduction of contralateral hand tremor up to 90 %. Fragile X–associated tremor/ataxia syndrome (FXTAS) is a progressive, late-onset neurodegenerative disorder associated with the FMR1 gene premutation. The treatment of FXTAS is challenging, and 6 patients with FXTAS who had tremor as the prevalent symptom have been successfully treated with VIM DBS. However, the worsening of ataxia has emerged as a concern with bilateral DBS procedures. These researchers described the 6-month follow-up of left Vim MRgFUS in an 82-year old man with long-standing genetically confirmed FXTAS (106 repeats of the FMR1 gene) complicated by disabling intention tremor and mild mid-line ataxia. The procedure (15 sonifications, average time of 13 seconds, power range of 150.0 to 725.0 W) was uneventful and caused a marked and immediate improvement of the contralateral tremor without worsening of the underlying ataxia. Post-operatively, these researchers examined the diffusion tensor imaging-based connectivity of the lesion with structural (3-D fast spoiled gradient echo T1, 2-D gradient echo) and diffusion-weighted (60 directions of diffusion gradients, field of view = 24; number of slices = 44; 1.8 ×1.8 × 2 mm voxel size; repetition time = minimum; b = 1,000 s/mm2; matrix = 128 × 128) images acquired on a 3-Tesla MRI scanner using part of a methodology previously reported. Briefly, raw diffusion images were corrected for distortion using BrainSuite software and then imported into StealthViz software for correction of motion artifacts and tensors calculation; the segmented region of interest from the lesion was then used as a seed for deterministic single-tensor tractography. The authors concluded that the safety and effectiveness of MRgFUS in ET and other tremor disorders as well as the clinical value of diffusion tensor imaging for VIM targeting need to be further explored. Moreover, they stated that although MRgFUS may be preferred over DBS in certain patient populations (particularly in older patients and those with brain atrophy similar to these patients), further research is also needed to compare its safety and effectiveness with that of DBS. This study provided Class IV evidence (single observational study without controls) that Vim MRgFUS might be safe and effective in patients with FXTAS.

Campins-Romeu et al (2024) noted that FXTAS is an adult-onset neurodegenerative disease caused by a premutation in the Fragile X mental retardation (FMR1) gene. For patients with limited daily-life activities due to medically intractable tremor, neurofunctional surgery approaches should be considered. Recent studies have reported the safety and effectiveness of unilateral MRIgFUS thalamotomy for patients with ET and tremor dominant PD. Currently, little knowledge is available when it comes to FXTAS patients and MRIgFUS. So far, only 2 FXTAS cases treated with unilateral MRIgFUS thalamotomy have been published; however, no long-term follow-up information is available. These researchers described the case of a 3-year follow-up of right VIM (ventralis intermediate nucleus) MRIgFUS thalamotomy in a left-handed patient with a confirmed FMR1 mutation. Due to the severe limitation in daily-life activities, neurofunctional surgical treatments were offered. The subject decided for MRIgFUS thalamotomy based on the limited experience previously published. MRIgFUS thalamotomy was accepted as compassionate use treatment by health authorities; and informed consent was signed by the patient. These investigators noted that it is well known that gait impairment is one of the most common complaint encountered during the first 3 months after surgery. Accordingly, in this case report, the decline noticed in the Berg Balance Score (BBS; a 14-item objective measure that assesses static balance and fall risk in adults) at the 1st month and its recovering at 1 year may probably be due to the perilesional edema; however, the mild worsening of balance at 3 years follow-up pointed to the intrinsic progression of the disease. The authors concluded that, to their knowledge, this was the 1st long-term reported follow-up of a FXTAS patient treated with unilateral MRIgFUS thalamotomy; therefore, it suggested acceptable safety and effectiveness of this technique when applied to FXTAS patients. Moreover, these researchers stated that the own progression of this neurodegenerative condition may negatively impact the future outcome of MRIgFUS, consequently patients should be evaluated and informed with caution. Equally, accurate surgical targeting and technique optimization are 2 important aspects of this novel technique that should be addressed.

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MRI-Guided Focused Ultrasound Thalamotomy for ET and Non-ET Tremor Syndromes

In an open-label, uncontrolled, pilot study, Elias et al (2013) investigated the use of transcranial MRI-guided focused ultrasound thalamotomy for the treatment of essential tremor. From February 2011 through December 2011, these researchers used transcranial MRI-guided focused ultrasound to target the unilateral ventral intermediate nucleus of the thalamus in 15 patients with severe, medication-refractory essential tremor. They recorded all safety data and measured the effectiveness of tremor suppression using the Clinical Rating Scale for Tremor to calculate the total score (ranging from 0 to 160), hand subscore (primary outcome, ranging from 0 to 32), and disability subscore (ranging from 0 to 32), with higher scores indicating worse tremor. These investigators assessed the patients' perceptions of treatment efficacy with the Quality of Life in Essential Tremor Questionnaire (ranging from 0 to 100 %, with higher scores indicating greater perceived disability). Thermal ablation of the thalamic target occurred in all patients. Adverse effects of the procedure included transient sensory, cerebellar, motor, and speech abnormalities, with persistent paresthesia in 4 patients. Scores for hand tremor improved from 20.4 at baseline to 5.2 at 12 months ($p = 0.001$). Total tremor scores improved from 54.9 to 24.3 ($p = 0.001$). Disability scores improved from 18.2 to 2.8 ($p = 0.001$). Quality-of-life scores improved from 37 % to 11 % ($p = 0.001$). The authors concluded that in this pilot study, essential tremor improved in 15 patients treated with MRI-guided focused ultrasound thalamotomy. They stated that large, RCTs are needed to assess the procedure's safety and effectiveness.

In a randomized controlled clinical study, Elias et al (2016) enrolled patients with moderate-to-severe essential tremor that had not responded to at least two trials of medical therapy and randomly assigned them in a 3:1 ratio to undergo unilateral focused ultrasound thalamotomy or a sham procedure. The Clinical Rating Scale for Tremor and the Quality of Life in Essential Tremor Questionnaire were administered at baseline and at 1, 3, 6, and 12 months. Tremor assessments were videotaped and rated by an independent group of neurologists who were unaware of the treatment assignments. The primary outcome was the between-group difference in the change from baseline to 3 months in hand tremor, rated on a 32-point scale (with higher scores indicating more severe tremor). After 3 months, patients in the sham-procedure group could cross over to active treatment (the open-label extension cohort). The investigators included 67 patients in the analysis. Hand-tremor scores improved more after focused ultrasound thalamotomy (from 18.1 points at baseline to 9.6 at 3 months) than after the sham procedure (from 16.0 to 15.8 points); the between-group difference in the mean change was 8.3 points (95% confidence interval [CI], 5.9 to 10.7; $P<0.001$). The improvement in the thalamotomy group was maintained at 12 months (change from baseline, 7.2 points; 95% CI, 6.1 to 8.3). Secondary outcome measures assessing disability and quality of life also improved with active treatment (the blinded thalamotomy cohort) as compared with the sham procedure ($P<0.001$ for both comparisons). Adverse events in the thalamotomy group included gait disturbance in 36% of patients and paresthesias or numbness in 38%; these adverse events persisted at 12 months in 9% and 14% of patients, respectively.

An editorial accompanying the study by Elias et al stated that the results are promising, particularly since this procedure, unlike traditional thalamotomy, does not require entering the cranium with a probe (Louis, 2016). The editorialist noted, however, that there are several important concerns about this study. The first is the limited follow-up period, which was 12 months. The sustained benefit at 2 years, 3 years, and 5 or more years is not known. The editorialist stated that studies with longer follow-up intervals are needed to address this issue, noting that this is particularly important because of tachyphylaxis, which is the second concern. The primary outcome measure was the score for hand tremor (on a scale ranging from 0 to 32, with higher scores indicating more severe tremor) at 3 months. The tremor score in the group that underwent focused ultrasound thalamotomy increased from 8.84 (at 1 month) to 9.55 (at 3 months) to 10.13 (at 6 months) to 10.89 (at 12 months). The increase from months 1 to 12 was 23%. Secondary outcome measures showed similar or greater increases during the 12-month follow-up period (e.g., the Clinical Rating Scale for Tremor score increased from 23.38 at 1 month to 32.38 at 12 months, which is a 38% increase). The editorialist stated that it is not clear whether this loss of efficacy, which is also seen to some extent with deep-brain stimulation, is due to disease progression or tolerance, although typical estimates of the rate of disease progression in essential tremor make the former possibility less likely. Third, the procedure did not achieve large improvements in everyone; the percentage change in tremor was less than 20% in 9 of 56 patients.

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The editorialist also noted several important caveats (Louis, 2016). The first is that the procedure is a thalamotomy. Hence, it creates a fixed brain lesion. With deep-brain stimulation, there is the potential to adjust stimulator settings in order to obtain further therapeutic gains, but there is no such potential with thalamotomy. Second, the procedure is not suitable for all patients; in particular, skull thickness presents a problem in some cases. Third, the most common side effect involved altered sensation, and this deficit remained permanent in 14% of patients. The editorialist suggested a head-to-head comparison with deep-brain stimulation to facilitate the direct comparison of the two approaches.

Bond and colleagues (2017) evaluated the safety and efficacy at 12-month follow-up, accounting for placebo response, of unilateral FUS thalamotomy for patients with tremor-dominant Parkinson disease (TDPD). Of the 326 patients identified from an in-house database, 53 patients consented to be screened; 26 were ineligible, and 27 were randomized (2:1) to FUS thalamotomy or a sham procedure at 2 centers from October18, 2012, to January 8, 2015. The most common reasons for disqualification were withdrawal (8 persons [31 %]), and not being medication refractory (8 persons [31 %]). Data were analyzed using intention-to-treat analysis, and assessments were double-blinded through the primary outcome. A total of 20 patients were randomized to unilateral FUS thalamotomy, and 7 to sham procedure. The sham group was offered open-label treatment after un-blinding. The pre-defined primary outcomes were safety and difference in improvement between groups at 3 months in the on-medication treated hand tremor sub-score from the CRST; secondary outcomes included descriptive results of UPDRS scores and quality of life (QOL) measures. Of the 27 patients, 26 (96 %) were men and the median age was 67.8 years (interquartile range [IQR], 62.1 to 73.8 years). On-medication median tremor scores improved 62 % (IQR, 22 % to 79 %) from a baseline of 17 points (IQR, 10.5 to 27.5) following FUS thalamotomy and 22 % (IQR, -11 % to 29 %) from a baseline of 23 points (IQR, 14.0 to 27.0) after sham procedures; the between-group difference was significant (Wilcoxon $p = 0.04$). On-medication median UPDRS motor scores improved 8 points (IQR, 0.5 to 11.0) from a baseline of 23 points (IQR, 15.5 to 34.0) following FUS thalamotomy and 1 point (IQR, -5.0 to 9.0) from a baseline of 25 points (IQR, 15.0 to 33.0) after sham procedures. Early in the study, heating of the internal capsule resulted in 2 cases (8 %) of mild hemiparesis, which improved and prompted monitoring of an additional axis during magnetic resonance thermometry. Other persistent AEs were orofacial paresthesia (4 events [20 %]), finger paresthesia (1 event [5 %]), and ataxia (1 event [5 %]). The authors concluded that preliminary results from this RCT on the efficacy of unilateral FUS thalamotomy for the treatment of patients with TDPD are encouraging. A notable placebo response was observed with sham procedures, necessitating a larger study to prove efficacy. Adverse events were similar to those of other thalamotomy procedures and will likely further improve as the technology for monitoring the FUS thalamotomy procedure improves.

The trial had several drawbacks -- small sample size ($n = 20$ in the FUS thalamotomy group), and the planned study enrollment of 30 patients was not reached. Medication dose was not fixed during the trial, potentially confounding the results. The trial was not designed to compare FUS thalamotomy with other treatments, such as DBS or gamma knife radiosurgery.

In a prospective, uncontrolled, single-center interventional study, Schreglmann and co-workers (2017) reported results of a prospective trial of unilateral transcranial MRgFUS ablation of the cerebello-thalamic tract (CTT) in ET. Motor symptoms, manual dexterity, cognition, and quality of life were assessed before intervention and at 48 hours and 1, 3, and 6 months after intervention. Rating of standardized video recordings was blinded for evaluation time points. Primary outcome was the change in unilateral hand tremor score of the treated hand. A total of 6 patients received MRgFUS ablation of the CTT contralateral to the treated hand. Repeated-measures comparison determined a statistically significant 83 % reduction (before versus 6 months after intervention mean ± SD; absolute reduction; 95 % CI) in the unilateral treated hand sub-score (14.3 ± 4.9 versus 2.5 ± 2.6; 11.8; 8.4 to 15.2; $p < 0.001$), while quality of life improved by 52 % (50.5 ± 19.4 versus 24.8 ± 11.4; 25.7; 3.5 to 47.28; $p = 0.046$). Measures for manual dexterity, attention and coordination, and overall cognition were unchanged. Transient side effects ($n = 3$) were ipsilateral hand clumsiness and mild gait instability for up to 3 months. The authors concluded that unilateral MRgFUS lesioning of the CTT was highly effective in reducing contralateral hand tremor in ET without affecting fine motor function and dexterity over 6 months of follow-up; adverse effects were mild and transient. Moreover, they stated that larger trials with a longer follow-up are needed to adequately evaluate the long-term safety and effectiveness of MRgFUS CTT ablation. This study provided Class IV evidence that for patients with ET, transcranial MRgFUS ablation of the CTT improved tremor.

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Fasano and colleagues (2017) reported the 6-month single-blinded results of unilateral thalamotomy with MRgFUS in patients with tremors other than ET. Three patients with tremor due to PD, 2 with dystonic tremor in the context of cervico-brachial dystonia and writer's cramp, and 1 with dystonia gene-associated tremor underwent MRgFUS targeting the ventro-intermedius nucleus (Vim) of the dominant hemisphere. The primary end-point was the reduction of lateralized items of the Tremor Rating Scale of contralateral hemi-body assessed by a blinded rater. All patients achieved a statistically significant, immediate, and sustained improvement of the contralateral tremor score by 42.2 %, 52.0 %, 55.9 %, and 52.9 % at 1 week and 1, 3, and 6 months after the procedure, respectively. All patients experienced transient side effects and 2 patients experienced persistent side effects at the time of last evaluation: hemi-tongue numbness and hemiparesis with hemi-hypoesthesia. The authors concluded that Vim MRgFUS is a promising, incision-free, but nevertheless invasive technique to treat tremors other than essential tremor. They stated that future studies on larger samples and longer follow-up are needed to further define its safety and effectiveness.

Chang and co-workers (2018) reported results of 2- year follow-up after MRgFUS thalamotomy for ET. A total of 76 patients with moderate-to-severe ET, who had not responded to at least 2 trials of medical therapy, were enrolled in the original randomized study of unilateral thalamotomy and evaluated using the clinical rating scale for tremor; 67 of the patients continued in the open-label extension phase of the study with monitoring for 2 years; 9 patients were excluded by 2 years (e.g., because of alternative therapy such as deep brain stimulation (n = 3) or inadequate thermal lesioning (n = 1)). However, all patients in each follow-up period were analyzed. Mean hand tremor score at baseline (19.8 ± 4.9, 76 patients) improved by 55 % at 6 months (8.6 ± 4.5, 75 patients). The improvement in tremor score from baseline was durable at 1 year (53 %, 8.9 ± 4.8, 70 patients) and at 2 years (56 %, 8.8 ± 5.0, 67 patients). Similarly, the disability score at baseline (16.4 ± 4.5, 76 patients) improved by 64 % at 6 months (5.4 ± 4.7, 75 patients). This improvement was also sustained at 1 year (5.4 ± 5.3, 70 patients) and at 2 years (6.5 ± 5.0, 67 patients). Paresthesias and gait disturbances were the most common AEs at 1 year -- each observed in 10 patients with an additional 5 patients experiencing neurological adverse effects. None of the AEs worsened over the period of follow-up and 2 of these resolved. There were no new delayed complications at 2 years. The authors concluded that tremor suppression after MRgFUS thalamotomy for ET was stably maintained at 2 years; latent or delayed complications did not develop after treatment. Moreover, they stated that additional follow-up is needed to determine the incidence of recurrence and the efficacy of MRgFUS over the long-term.

The authors stated that there were some important limitations as well as differences and discrepancies between these findings and the previous report on this cohort of treated patients. First, the number of reported patients was different; 56 subjects in the previous report were compared to the 20 sham-treated patients, but here we analyzed all 76 patients. Second, the follow-up was versus 2 years here versus 1 year previously. Third, as is characteristic with increasing duration of follow-up, there were patients who dropped-out of the study by 2 years. In total, 9 patients, many of whom had unsuccessful treatment or suboptimal benefit, crossed-over to an alternative treatment or dropped-out or were lost to follow-up at 2 years. The exclusion of non-responders from the analysis introduced a bias and over-estimated the benefit in those patients that remained in the study.

Zaaroor and associates (2018) noted that thalamotomy of the Vim is effective in alleviating medication-resistant tremor in patients with ET and PD; MRgFUS is an innovative technology that enables non-invasive thalamotomy via thermal ablation. Patients with severe medication-resistant tremor underwent unilateral Vim thalamotomy using MRgFUS. Effects on tremor were evaluated using the CRST in patients with ET and by the motor part of the UPDRS in patients with PD and ET-PD (defined as patients with ET who developed PD many years later). Quality of life in ET was measured by the Quality of Life in Essential Tremor (QUEST) questionnaire and in PD by the PD Questionnaire (PDQ-39). A total of 30 patients underwent MRgFUS, including 18 with ET, 9 with PD, and 3 with ET-PD. The mean age of the study population was 68.9 ± 8.3 years (range of 46 to 87 years) with a mean disease duration of 12.1 ± 8.9 years (range of 2 to 30 years). MRgFUS created a lesion at the planned target in all patients, resulting in cessation of tremor in the treated hand immediately following treatment. At 1 month post-treatment, the mean CRST score of the patients with ET decreased from 40.7 ± 11.6 to 9.3 ± 7.1 ($p < 0.001$) and was 8.2 ± 5.0 at 6 months after treatment ($p < 0.001$, compared with baseline). Average QUEST scores decreased from 44.8 ± 12.9 to 13.1 ± 13.2 ($p < 0.001$) and was 12.3 ± 7.2 at 6 months after treatment ($p < 0.001$). In patients with PD, the mean score of the motor part of the UPDRS decreased from 24.9 ± 8.0 to 16.4 ± 11.1 ($p = 0.042$) at 1

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month and was 13.4 ± 9.2 at 6 months after treatment (p = 0.009, compared with baseline). The mean PDQ-39 score decreased from 38.6 ± 16.8 to 26.1 ± 7.2 (p = 0.036) and was 20.6 ± 8.8 at 6 months after treatment (p = 0.008). During follow-up of 6 to 24 months (mean of 11.5 ± 7.2 months, median of 12.0 months), tremor re-appeared in 6 of the patients (2 with ET, 2 with PD, and 2 with ET-PD), to a lesser degree than before the procedure in 5; AEs that transiently occurred during sonication included headache (n = 11), short-lasting vertigo (n = 14) and dizziness (n = 4), nausea (n = 3), burning scalp sensation (n = 3), vomiting (n = 2) and lip paresthesia (n = 2); AEs that lasted after the procedure included gait ataxia (n = 5), unsteady feeling (n = 4), taste disturbances (n = 4), asthenia (n = 4), and hand ataxia (n = 3). No AE lasted beyond 3 months. Patients underwent on average 21.0 ± 6.9 sonications (range of 14 to 45 sonications) with an average maximal sonication time of 16.0 ± 3.0 seconds (range of 13 to 24 seconds). The mean maximal energy reached was 12,500 ± 4,274 J (range of 5,850 to 23,040 J) with a mean maximal temperature of 56.5° ± 2.2° C (range of 55° to 60° C). The authors concluded that MRgFUS Vim thalamotomy to relieve medication-resistant tremor was safe and effective in patients with ET, PD, and ET-PD; current results emphasized the superior AEs profile of MRgFUS over other surgical approaches for treating tremor with similar efficacy. Moreover, they stated that large randomized studies are needed to evaluate long-term safety and effectiveness.

Rohani and Fasano (2017) noted that while there is no breakthrough progress in the medical treatment of ET, in the past decades several remarkable achievements happened in the surgical field, such as radiofrequency (RF) thalamotomy, thalamic DBS, and gamma knife thalamotomy. The most recent advance in this area is MRgFUS. These researchers discussed the new developments and trials of MRgFUS in the treatment of ET and other tremor disorders. MRgFUS is an incisionless surgery performed without anesthesia and ionizing radiation (no risk of cumulative dose and delayed side effects). Studies have shown the safety and effectiveness of unilateral MRgFUS-thalamotomy in the treatment of ET. It has been successfully used in a few patients with PD-related tremor, and in fewer patients with fragile X-associated tremor/ataxia syndrome. The safety and long-term effects of the procedure are still unclear, as temporary and permanent AEs have been reported as well as recurrence of tremor. The authors concluded that MRgFUS is a promising new surgical approach with a number of unknowns and unsolved issues. It represents a valuable option particularly for patients who refused or could not be candidates for other procedures, DBS in particular.

These investigators noted that MRgFUS has a number of problems/unknowns:

Problems

- Variable effects on symptoms control
- Decay of tremor control in the short-term
- Relatively high number of persistent side effects
- Unpredictable hyper-response of brain tissue
- Not suitable to target both hemispheres
- Not possible in patients with MRI contraindications
- Not possible in patients with high skull thickness
- Not possible in patients with previous brain surgery
- Patients' misperception of being non-surgical

Unknowns

- Long-term effects
- Re-operation of the same brain area (e.g., in case of tremor recurrence)
- Efficacy of lesioning less centered brain targets (e.g., GPi)
- Safety of bilateral procedures
- Efficacy of DTI MRI to better target brain nuclei/fibers
- Safety of STN lesioning (risk of hemiballismus)
- Bleeding risk in selected populations (e.g., patients on anticoagulants)
- Impact of placebo effect in previous and future RCTs

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These researchers stated that MRgFUS is perceived as a safe procedure but it has been associated with a relatively high number of persistent side effects: 9 % of gait disturbance and 14 % of paresthesia 1 year after surgery in a large trial. MRgFUS-thalamotomy is created without electrophysiological localization techniques that were developed for RF thalamotomy (intra-operative recording and stimulation). Given the fact that the main danger posed by RF-thalamotomy is not the incision, burr hole, or electrode pass but the ablation itself, it has been recently commented that "MRgFUS thalamotomy may actually be riskier than classic RF-thalamotomy, which, in turn, is riskier than DBS". On the other hand, during the MRgFUS procedure real-time brain MRI is used to monitor target localization and the size of the ablation area. Another still not fully elucidated problem is the unpredictable brain tissue reaction seen in some patients, a phenomenon already reported in GK procedures. It consists of a large amount of edema and non-radial spreading of lesioning effects. The former is typically associated with transient adverse effects; the latter is more dangerous and its phenomenology probably relies on the anatomy of the targeted area (fiber directions and ratio between neuronal bodies and axons). Another problem of MRgFUS for movement disorders is related to the fact that many patients need bilateral procedures and this limits in particular the usefulness of this technique in pallidotomies performed to treat diseases such as dystonia or PD. In addition, it is not emphasized enough that many patients cannot undergo MRgFUS; examples include patients with pace-makers or other contraindications to MRI or patients with high skull thickness. In fact, a study on 25 patients (1 with PD, 15 with ET, and 9 with OCD) found that skull volume and density significantly affect the maximum temperature achieved in the deep brain. For example, patients with high skull density may undergo skin lesions and local pain in order to receive enough lesioning energy in the deep brain. Another contraindication of MRgFUS is a history of previous brain surgery, because some of these patients' brain areas may receive more energy than predicted by software, assuming that the entire skull is intact. Probably the most important hurdle of MRgFUS is related to the many unknowns of a relatively new treatment that gained popularity rather fast. In this respect, although public opinion and patients perceive it as a "non-surgical" intervention, MRgFUS is not risk-free. This virtually puts many people at risk, as ET is the most prevalent movement disorder and up to 50 % of patients become refractory or intolerant to medication.

The authors stated that this review has emphasized the many problems and unknowns related to this novel procedure. It is too early to draw definite conclusions on the value and unsolved issues of MRgFUS, but the good news is that one more option is now available for tremor patients. They believe that a deep understanding of the safety and efficacy of these procedures is needed for the appropriate selection of the surgical patients; future studies comparing the different treatment modalities are certainly needed.

In a prospective, uncontrolled, single-center interventional study, Schreglmann and associates (2018) reported results of a prospective trial of unilateral transcranial MRI-guided focused ultrasound (MRIgFUS) ablation of the cerebello-thalamic tract (CTT) in ET. Patients with ET fulfilling criteria for interventional therapy received unilateral ablation of the CTT by MRIgFUS. Motor symptoms, manual dexterity, cognition, and QOL were assessed before intervention and at 48 hours and 1, 3, and 6 months after intervention. Rating of standardized video recordings was blinded for evaluation time-points. Primary outcome was the change in unilateral hand tremor score of the treated hand. A total of 6 patients received MRIgFUS ablation of the CTT contralateral to the treated hand. Repeated-measures comparison determined a statistically significant 83 % reduction (before versus 6 months after intervention mean ± SD; absolute reduction; 95 % CI) in the unilateral treated hand sub-score (14.3 ± 4.9 versus 2.5 ± 2.6; 11.8; 8.4 to 15.2; $p < 0.001$), while QOL improved by 52 % (50.5 ± 19.4 versus 24.8 ± 11.4; 25.7; 3.5 to 47.28; $p = 0.046$). Measures for manual dexterity, attention and coordination, and overall cognition were unchanged. Transient side effects ($n = 3$) were ipsilateral hand clumsiness and mild gait instability for up to 3 months. The authors concluded that unilateral MRIgFUS lesioning of the CTT was highly effective in reducing contralateral hand tremor in ET without affecting fine motor function and dexterity over 6 months of follow-up. Adverse effects were mild and transient. Level of Evidence = IV. The authors stated that this was a proof of principle study; and larger trials with a longer follow-up are needed to adequately evaluate the long-term safety and efficacy of MRIgFUS CTT ablation.

Mohammed and colleagues (2018) noted that MRgFUS is a novel technique that uses high-intensity focused ultrasound (HIFU) to achieve target ablation. Like a lens focusing the sun's rays, the US waves are focused to generate heat. This therapy combines the non-invasiveness of Gamma Knife thalamotomy and the real-time ablation of DBS with acceptable complication rates. These investigators analyzed the overall outcomes

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and complications of MRgFUS in the treatment of ET. They carried out a meta-analysis in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines by searching PubMed, Cochrane library database, Web of Science, and Cumulative Index to Nursing and Allied Health Literature (CINAHL). Patients with the diagnosis of ET who were treated with MRgFUS were included in the study. The change in the CRST score after treatment was analyzed. The improvement in disability was assessed with the QUEST score. The pooled data were analyzed by the DerSimonian-Laird random-effects model. Tests for bias and heterogeneity were performed. A total of 9 studies with 160 patients who had ET were included in the meta-analysis. The ventral intermediate nucleus was the target in 8 of the studies. The cerebello-thalamic tract was targeted in 1 study. There was 1 RCT, 6 studies were retrospective, and 2 were prospective. The mean number of sonications given in various studies ranged from 11 ± 3.2 to 22.5 ± 7.5 (mean ± SD). The maximum delivered energy ranged from 10,320 ± 4,537 to 14,497 ± 6,695 Joules. The mean of peak temperature reached ranged from 53° C ± 2.3° C to 62.0° C ± 2.5° C. On meta-analysis with the random-effects model, the pooled percentage improvements in the CRST Total, CRST Part A, CRST Part C, and QUEST scores were 62.2 %, 62.4 %, 69.1 %, and 46.5 %, respectively. Dizziness was the most common in-procedure complication, occurring in 45.5 %, followed by nausea and vomiting in 26.85 % (pooled percentage). At 3 months, ataxia was the most common complication, occurring in 32.8 %, followed by paresthesia in 25.1 % of the patients. At 12 months post-treatment, the ataxia had significantly recovered and paresthesia became the most common persisting complication, at 15.3 %. The authors concluded that MRgFUS therapy for ET significantly improved the CRST scores and improved the QOL in patients with ET, with an acceptable complication rate. They stated that therapy with MRgFUS is a promising frontier in functional neurosurgery.

Langford and co-workers (2018) stated that up to 50 % of ET patients are refractory to medication and require alternative treatment to achieve tremor relief. These researchers analyzed evidence supporting the use of the emerging MRgFUS compared to alternative stimulatory and ablative interventions for the treatment of medication-refractory ET: RF thalamotomy, unilateral DBS, and SRS. They performed a systematic literature review to identify clinical, health-related QOL (HRQOL), and economic evidence for each intervention. Because of the lack of comparative evidence captured, a feasibility assessment was performed to determine possible comparisons between interventions, and newly established matching-adjusted indirect comparison and simulated treatment comparison techniques were used to conduct a comparison between unilateral DBS aggregate data and MRgFUS individual patient data. The systematic literature review identified 1,559 records, and screening yielded 46 relevant articles. The captured studies demonstrated that RF thalamotomy, DBS, SRS, and MRgFUS all exhibit clinical efficacy, with variation in onset and duration of tremor relief, and are each associated with a unique safety profile. The matching-adjusted indirect comparison and simulated treatment comparison results demonstrated no evidence of a difference in efficacy (measured by Clinical Rating Scale for Tremor Total) and HRQOL (measured by CRST Part C) outcomes between MRgFUS and unilateral DBS in the short-term (less than or equal to 12 months). The authors concluded that the findings of this study provided preliminary evidence that MRgFUS could elicit similar short-term tremor- and HRQOL-related benefits to DBS, the current standard of care, and allowed for the first robust statistical comparison between these interventions.

Halpern and colleagues (2019) tested the hypothesis that transcranial MRgFUS (tcMRgFUS) thalamotomy is effective, durable, and safe for patients with medication-refractory ET. These researchers examined clinical outcomes at 3-year follow-up of a controlled multi-center prospective trial. Outcomes were based on the Clinical Rating Scale for Tremor, including hand combined tremor-motor (scale of 0 to 32), functional disability (scale of 0 to 32), and postural tremor (scale of 0 to 4) scores, and total scores from the QOL in Essential Tremor Questionnaire (scale of 0 to 100). Scores at 36 months were compared with baseline and at 6 months after treatment to evaluate the durability and effectiveness; AEs were also reported. Measured scores remained improved from baseline to 36 months (all $p < 0.0001$). Range of improvement from baseline was 38 % to 50 % in hand tremor, 43 % to 56 % in disability, 50 % to 75 % in postural tremor, and 27 % to 42 % in QOL. When compared to scores at 6 months, median scores increased for hand tremor (95 % CI: 0 to 2, $p = 0.0098$) and disability (95 % CI: 1 to 4, $p = 0.0001$). During the 3rd year follow-up, all previously noted AEs remained mild or moderate, none worsened, 2 resolved, and no AEs occurred. The authors concluded that results at 3 years following unilateral tcMRgFUS thalamotomy for ET showed continued benefit, and no progressive or delayed complications. Patients may experience mild degradation in some treatment metrics by 3 years, though improvement from baseline remained significant. This study

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provided Class IV evidence that for patients with severe ET, unilateral tcMRgFUS thalamotomy provided durable benefit after 3 years. It should be noted that this trial had a 31 % withdrawal rate at the 36-month time-point, which could have biased the final findings. Furthermore, this study was partially funded by InSightec, and some authors received funding from InSightec.

The authors stated that this study had 2 main drawbacks. First, the patient drop-out rate in this trial was 31 % in 3 years. At 3 months after treatment, these patients had significantly different tremor reduction than the patients who stayed in the trial, implying that the differences these researchers measured compared to baseline would be less if these missing patients had remained in the study. Nevertheless, this analysis showed that even if these investigators assumed that all of the missing patients lost all benefit of treatment and returned to their baseline tremor scores by 3 years, the overall cohort would still have significantly improved tremor scores compared to baseline. It should be noted that only 4 patients (5 %) pursued alternative intervention for their tremor, suggesting that the sensitivity analysis approach was conservative. Second, these investigators analyzed a patient cohort that was different from the cohorts presented in the original RCT or analyzed at the 2-year follow-up. Unlike the RCT, these researchers included in the same analysis the patients from the original treatment arm and those who crossed-over, and differently from the 2-year follow-up study, these researchers included in the primary analysis only the observed data of patients who presented to the 3-year study visit. Tremor is expected to worsen over time, so carrying forward prior scores could have optimistically biased the results. The authors therefore also used imputed values for missing data based on a conservative assumption that all treatment benefit had been lost. These approaches all indicated a significant improvement in tremor and QOL with tcMRgFUS thalamotomy. This method also provided the expected range of responses 3 years after tcMRgFUS treatment.

Park et al (2019) noted that following the emergence of magnetic resonance-guided focused ultrasound (MRgFUS) as a promising tool for movement disorder surgery, thalamotomy for essential tremor using this technique has become a useful tool based on its efficacy and lack of adverse effects. These researchers summarized the 4-year results of previous reports focusing on the durability of effectiveness of MRgFUS thalamotomy for essential tremor (ET). From October 2013 to August 2014, a total of 15 patients with intractable ET were enrolled; 12 of them completed clinical assessment through 4 years of post-operative follow-up. Tremor severity, task performance, and disability were measured using the Clinical Rating Scale of Tremor. The mean age of the 12 patients was 61.7 ± 8.1 years. Maximally delivered energy was 15,552.4 ± 6,574.1 joules. The mean number of sonications was 17.3 ± 1.6. The mean post-operative lesion volume was 82.6 ± 29.023 mm3 and in 1 year was a mean of 9.667 ± 8.573 mm3 . Four years post-operatively, improvement of the hand tremor score was 56 %, that of the disability score was 63 %, that of the postural score was 70 %, and that of the action score was 63 % compared with baseline; all improvements were significant and sustained over the 4-year period after thalamotomy. There was no permanent adverse effect throughout the 4-year follow-up period. The authors concluded that MRgFUS thalamotomy exhibited sustained clinical efficacy 4 years following the treatment of intractable ET; AEs were generally transient. These researchers stated that a large cohort of patients who have undergone MRgFUS thalamotomy with longer follow-up is needed to confirm these findings. Limitations of the study were the small number of subjects that completed four-year followup.

Harary and associates (2019) noted that the predominant neurosurgical approach to medication-refractory ET is thalamic DBS. The emergence of MRgFUS thalamotomy has re-awakened the debate surrounding the use of DBS versus thalamotomy for this indication. These researchers provided a contemporary comparison between DBS and MRgFUS. Two controlled trials that evaluated DBS and MRgFUS for the unilateral treatment of refractory ET were compared. Clinical outcomes extracted included postural tremor score in the treated upper extremity, QOL, and incidence of adverse events (AE). Baseline patient characteristics were comparable in the 2 studies, except that DBS patients were younger and had more severe baseline tremor. Both DBS- and MRgFUS-treated patients had significant tremor improvement that was sustained for 1-year post-treatment, and significant improvement in QOL. The MRgFUS cohort had higher rates of persistent neurologic AE, whereas the DBS group had higher rates of surgery- and hardware-related AEs, including intra-cranial hemorrhage. The authors concluded that in context of prior literature, both DBS and MRgFUS significantly improved tremor control and QOL. The 2 approaches were predominantly differentiated by their AE-profile. Moreover, these researchers stated that additional head-to-head comparison on matched clinical populations are needed to better compare clinical efficacy and long-term outcomes.

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Altinel et al (2019) conducted a meta-analysis of randomized clinical trials of patients with tremor treated with either lesion surgery (LS), DBS, or controls. The outcomes were the change in tremor score, QOL, cognitive function, and neuropsychiatric function. A total of 15 trials, including 1508 patients, met eligibility criteria. These researchers observed no significant difference in change of tremor scale (SMD L0.07, 95 % CI: 0.38 to 0.24), QOL (SMD 0.21, 95 % CI: 0.69 to 0.27), cognitive function (SMD 0.06, 95 % CI: 0.27 to 0.39), or neuropsychiatric function (SMD L0.15, 95 % CI: 0.49 to 0.19) between LS and stimulation surgery. These investigators observed heterogeneity across studies during indirect comparison of QOL.
They identified a possible effect modifier: improvement in QOL correlated with duration of disease ($p < 0.035$). The authors found that focused-ultrasound LS was associated with a 0.70 SMD increase ($p < 0.014$) in QOL versus DBS in an exploratory subgroup analysis by separating 2 studies with focused-ultrasound LS from other LS studies. The authors concluded that policy makers, health care providers, and patients could consider focused ultrasound LS as a potential choice for tremor control, based on currently available evidence; however, additional evidence from randomized trials comparing stimulation with the focused-ultrasound approach is needed given the lack of direct comparison between the two in the literature and therefore in this meta-analysis.

Giordano and colleagues (2020) stated that the current gold standard surgical treatment for medication-resistant ET is DBS; however, recent advances in technologies have led to the development of incisionless techniques, such as MRgFUS thalamotomy. These investigators carried out a systematic review according to the PRISMA statement to compare unilateral MRgFUS thalamotomy to unilateral and bilateral DBS in the treatment of ET in terms of tremor severity and QOL improvement. PubMed, Cochrane Database of Systematic Reviews, Cochrane Central Register of Controlled Trials and SCOPUS databases were searched. A total of 45 eligible articles, published between 1990 and 2019, were retrieved; 1,202 patients were treated with DBS and 477 were treated with MRgFUS thalamotomy. Post-operative tremor improvement was greater following DBS than MRgFUS thalamotomy ($p < 0.001$). A subgroup analysis was preformed stratifying by treatment laterality: bilateral DBS was significantly superior to both MRgFUS and unilateral DBS ($p < 0.001$), but no significant difference was observed between MRgFUS and unilateral DBS ($p < 0.198$). Post-operative QOL improvement was significantly greater following MRgFUS thalamotomy than DBS ($p < 0.001$). Complications were differently distributed among the 2 groups ($p < 0.001$). Persistent complications were significantly more common in the MRgFUS group ($p = 0.042$). The authors concluded that while bilateral DBS proved superior to unilateral MRgFUS thalamotomy in the treatment of ET, a subgroup analysis suggested that treatment laterality was the most significant determinant of tremor improvement, thus highlighting the importance of future investigations on bilateral staged MRgFUS thalamotomy.

The American Society of Stereotactic and Functional Neurosurgery (ASSFN), which acts as the joint section representing the field of stereotactic and functional neurosurgery on behalf of the Congress of Neurological Surgeons and the American Association of Neurological Surgeons, provided the expert consensus opinion on evidence-based best practices for the use and implementation of MRI-guided focused ultrasound thalamotomy for the management of ET. Based on current evidence, indications for treatment were confirmed diagnosis of ET, failure to respond to 1st-line therapies, disabling appendicular tremor, and unilateral treatment (Pouratian et al, 2020).

Bilateral-Staged Magnetic Resonance-Guided Focused Ultrasound Thalamotomy for the Treatment of Essential Tremor

Martinez-Fernandez and colleagues (2021) stated that unilateral MR-guided focused ultrasound (FUS) thalamotomy is effective for the treatment of medically refractory ET; however, viability of bilateral FUS ablation is unexplored. In a case-series study, patients diagnosed with medically refractory ET and previously treated with unilateral FUS thalamotomy at least 5 months before underwent bilateral treatment. The time-points were baseline (before 1st thalamotomy) and FUS1 and FUS2 (4 weeks before and 6 months after 2nd thalamotomy, respectively). The primary endpoint was safety. Effectiveness was evaluated via the CRST, which includes subscales for tremor examination (part A), task performance (part B) and tremor-related disability (part C). A total of 9 patients were treated; no permanent AEs were reported; 6 patients presented mild gait instability and 1 dysarthria, all resolving within the first few weeks; 3 patients reported

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peri-oral hypoesthesia, resolving in 1 case. Total CRST score improved by 71 % from baseline to FUS2 (from 52.3 ± 12 to 15.5 ± 9.4, $p < 0.001$), conveying a 67 % reduction in bilateral upper limb A+B (from 32.3 ± 7.8 to 10.8 ± 7.3, $p = 0.001$). Part C decreased by 81 % (from 16.4 ± 3.6 to 3.1 ± 2.9, $p < 0.001$). Reduction in head and voice tremor was 66 % (from 1.2 ± 0.44 to 0.4 ± 0.54, $p = 0.01$) and 45 % (from 1.8 ± 1.1 to 1 ± 0.8, $p = 0.02$), respectively. The authors concluded that bilateral staged FUS thalamotomy for ET was feasible and might be safe and effective; voice and head tremor might also improve. These researchers stated that a controlled trial is needed.

Focused Ultrasound Thalamotomy for the Treatment of Parkinson Disease

In a double-blind, sham-controlled, randomized clinical trial, Sperling and associates (2018) examined non-motor outcomes and correlates of QOL 3 and 12 months following unilateral focused US thalamotomy in tremor-dominant PD (TDPD). A total of 27 patients with TDPD underwent comprehensive neuropsychological evaluations. These included assessment of mood, behavior, and QOL at baseline, 3 months, 3 months after cross-over in the sham group, and 12 months after active treatment. These researchers used Mann-Whitney U tests to examine differences between the active (n = 20) and sham (n = 7) groups at 3 months and Friedman tests to evaluate within-group changes after active treatment. They assessed correlations between disease variables and post-operative QOL using Kendall tau-b tests. There were no differences in cognition, mood, or behavior between the active and sham groups at 3-month blinded assessment. After active treatment, there were no differences in mood or behavior. Only declines in Stroop Color Naming and phonemic fluency were observed. Patients experienced post-operative improvements in QoL and activities of daily living (ADL). Mood and behavioral symptoms, aspects of cognitive functioning, ADL, and overall motor symptom severity, but not tremor severity specifically, were associated with QOL. The authors concluded that in TDPD, unilateral focused US thalamotomy appeared safe from a cognitive, mood, and behavioral perspective; QOL and ADL significantly improved following surgery. Non-motor symptoms and ADL were more closely associated with QOL than tremor severity. This study provided Class II evidence that for patients with TDPD, unilateral focused US thalamotomy did not adversely change cognition, mood, or behavior at 3 months. The main drawbacks of this study were its small sample size (n = 20 in the treatment group) and medication dosages were not fixed during the trial. It was possible that the surgical procedure contributed to subacute non-motor changes that were subtle and no longer present at the 3-month neuropsychological follow-up. However, patients did not report adverse cognitive, mood, or behavioral changes during this subacute phase to suggest this to be true. The attrition rate at 12 months also limited interpretation of the longer-term follow-up results. However, there were no differences in motor or non-motor symptoms between the patients who were ultimately lost to 12-month follow-up and those who maintained participation. This suggested that the 12-month results were not affected by attrition.

Moosa and colleagues (2019) MR-guided focused US is a novel, minimally invasive surgical procedure for symptomatic treatment of Parkinson' disease (PD). With this technology, the ventral intermediate nucleus (VIN), sub-thalamic nucleus (STN), and internal globus pallidus (IGP) have been targeted for therapeutic cerebral ablation, while also minimizing the risk of hemorrhage and infection from more invasive neurosurgical procedures. The authors noted that, in a double-blinded, prospective, sham-controlled RCT of MR-guided focused US thalamotomy for treatment of tremor-dominant PD, 62 % of treated patients demonstrated improvement in tremor scores from baseline to 3 months post-operatively, as compared to 22 % in the sham group. The authors observed that there has been only 1 open-label trial of MR-guided focused US sub-thalamotomy for patients with PD, demonstrating improvements of 71 % for rigidity, 36 % for akinesia, and 77 % for tremor 6 months after treatment. Among the 2 open-label trials of MR-guided focused US pallidotomy for patients with PD, dyskinesia and overall motor scores improved up to 52 % and 45 % at 6 months post-operatively. The authors stated that, although MR-guided focused US thalamotomy is now approved by the U.S. Food and Drug Administration for treatment of parkinsonian tremor, additional high-quality RCTs are needed and are underway to determine the safety and efficacy of MR-guided focused US sub-thalamotomy and pallidotomy for treatment of the cardinal features of PD. The authors stated that these findings will be paramount to aid clinicians in determining the ideal ablative target for individual patients. These authors stated that additional work will be needed to examine the durability of MR-guided focused US lesions, ideal timing of MR-guided focused US ablation in the course of PD, and the safety of performing bilateral lesions.

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Focused Ultrasound Thalamotomy for Multiple Sclerosis-Associated Tremor

Manez-Miro and colleagues (2020) stated that multiple sclerosis (MS)-related tremor is frequent and can often be refractory to medical treatment, which makes it a potential source of major disability. Functional neurosurgery approaches such as thalamic DBS or RF thalamotomy are proven to be effective, but the application of invasive techniques in MS tremor has so far been limited. MRgFUS thalamotomy, which has already been approved for treating ET and parkinsonian tremor, provides a minimally invasive approach that could be useful in the management of MS tremor. The authors reported for the first time a patient with medically refractory MS-associated tremor successfully treated by FUS thalamotomy. These preliminary findings need to be validated by well-designed studies.

Ventro-Oral Thalamotomy for Focal Hand Dystonia

Horisawa and colleagues (2019) reported the safety and long-term efficacy of ventro-oral thalamotomy (VOT) for 171 consecutive patients with task-specific focal hand dystonia (TSFHD). Between October 2003 and February 2017, a total of 171 consecutive patients with TSFHD underwent unilateral VOT. Etiologies included writer's cramp (n = 92), musician's dystonia (n = 58), and other occupational task-related dystonia (n = 21). The TSFHD scale was used to evaluate patients' neurologic conditions (range of 1 to 5, high score indicated a better condition). The scores before surgery; at 1 week, 3 months, and 12 months post-operatively; and the last available follow-up period were determined. Post-operative complications and post-operative recurrence were also evaluated. The scores before surgery; at 1 week (1.72 ± 0.57, 4.33 ± 0.85 [$p < 0.001$]), 3 months (4.30 ± 1.06 [$p < 0.001$]), and 12 months (4.30 ± 1.13 [$p < 0.001$]); and the last available follow-up (4.39 ± 1.07 [$p < 0.001$]) post-operatively improved. The mean clinical follow-up period was 25.4 ± 32.1 months (range of 3 to 165). Permanent AEs developed in 6 patients (3.5 %); 18 patients developed recurrent dystonic symptoms post-operatively. Of these 18 patients, 9 underwent VOT again, of which 7 achieved improvement. The authors concluded that VOT was a feasible and reasonable treatment for patients with refractory TSFHD. These researchers stated that further well-powered studies (prospective, randomized, and double-blind) are needed to clarify more accurate assessment of the safety and efficacy of VOT for TSFHD. This study provided Class IV evidence. This was a retrospective, single-center, case-series study with no control group or blinding, which limited the generalizability of these findings.

Furthermore, an UpToDate review on “Treatment of dystonia” (Comella, 2019) does not mention ventro-oral thalamotomy as a therapeutic option for focal dystonia.

Intractable Pain

de la Torre and associates (2021) noted that phantom limb syndrome is defined as the perception of intense pain or other sensations that are secondary to a neural lesion in a limb that does not exist. It can be treated using pharmacological and surgical interventions. Most medications are prescribed to improve patients' lives; however, the response rate is low. These investigators presented a case of phantom limb syndrome in a 42-year-old woman with a history of trans-radial amputation of the left thoracic limb due to an accidental compression 1 year before. The patient underwent placement of a deep brain stimulator at the ventral posteromedial nucleus (VPM) on the right side and removal secondary to loss of battery. The patient continued to have a burning pain throughout the limb with a sensation of still having the limb, which was subsequently diagnosed as phantom limb syndrome. After a thorough discussion with the patient, a right stereotactic centro-median thalamotomy was offered. An immediate response was reported with a reduction in pain severity on the visual analog scale (VAS) from a pre-operative value of 9 to 10 to a post-operative value of 2, with no complications. The authors concluded that although phantom limb pain is one of the most difficult-to-treat conditions, centro-median thalamotomy may provide an effective treatment with adequate outcomes.

Stereotactic Radiofrequency Thalamotomy for the Treatment of Malignant Pain

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Haddad et al (2021) noted that pain is a common occurrence in patients with cancer, which, in some cases, is not adequately controlled with medical analgesia. Thalamotomy is a therapeutic option in such circumstances, but synthesis of historical evidence and thalamic stratified data are lacking. In a systematic review, these investigators examined evidence supporting RF thalamotomy for the treatment of intractable cancer pain. This review was carried out using multiple electronic databases and a (PICO) patient/problem, intervention, comparison, outcome search with the terms "radiofrequency thalamotomy" and "cancer pain". Of the 22 full-text studies examined for eligibility, 14 were included for review. Studies were excluded in which RF ablation (RFA) was not used, chronic implantation was used, or the study did not include patients with cancer pain. A total of 13 case series and 1 case report were included. Thalamic targets included ventral posterior, central lateral, dorsomedial, centro-median, centro-median/para-fascicular, centro-median and anterior pulvinar, pulvinar, limitans, supra-geniculate and posterior nuclei. Patient characteristics, operative methods, lesioning parameters, patient follow-up, and outcomes were variably reported across the studies. Where relevant outcome data were available, 97 % of patients experienced initial pain relief and 79 % experienced significant lasting relief; AEs were typically transient. The authors concluded that these limited findings showed that RF thalamotomy for cancer pain was well-tolerated and could produce significant relief from intractable cancer pain. No superiority of thalamic target could be determined. These investigators hoped that this review would serve as a stimulus to drive the consideration of thalamotomy as an effective palliative option for cancer pain and encourages the development of standardized protocols. Moreover, these researchers stated that the heterogeneous data regarding patient selection, follow-up, target choice, tumor characteristics, and comparable outcome reporting emphasize the need for larger studies that use homogenous populations with standardized outcome measures.

Farrell et al (2021) stated that cancer pain is common and challenging to manage. It is estimated that about 30 % of cancer patients have pain that is inadequately controlled by analgesia. These investigators discussed safe and effective neuro-ablative therapeutic options for refractory cancer pain. Current management of cancer pain predominantly focuses on the use of medications, resulting in a relative loss of knowledge of these surgical techniques and the erosion of the skills required to perform them. The authors reviewed surgical methods of modulating various points of the neural axis with the aim to expand the knowledge base of those managing cancer pain. Integration of neuro-ablative approaches may lead to higher rates of pain relief, and the opportunity to dose reduce analgesic agents with potential deleterious side effects. With an ever-increasing population of cancer patients, it is essential that neurosurgeons maintain or train in these techniques (including cingulotomy, cordotomy, myelotomy, and thalamotomy) in tandem with the oncological multi-disciplinary team.

These researchers stated that judicious use of any therapy is key to its success. It could be challenging to decide when a patient is an appropriate candidate for any of the neurosurgical interventions offered. What exactly is intractable pain and when does the patient qualify for the surgery? What therapeutic approaches must they have tried first? Given the patients are often at the end-stage of their disease process, how does this effect their performance status and fitness for surgery? These investigators noted that general consensus is to use the following selection criteria, which emphasize 5 domains that must be fulfilled before consideration for an ablative procedure: First, advanced oncological disease with limited life expectancy. Second, options for radiotherapy have been exhausted. Third, best medical treatment has failed to cause pain relief or incurred the development of intolerable side effects. Fourth, pain has failed to respond adequately to any targeted interventions (e.g.,, nerve blocks). Fifth, absence of technical limitations or medical contraindications to the procedures offered.

Franzini and colleagues (2022) stated that medial thalamotomy using SRS is a potential treatment for intractable pain; however, the ideal treatment parameters and expected outcomes from this procedure remain unclear. These investigators examined medial thalamotomy using SRS, specifically for intractable pain. They carried out a systematic review to identify all clinical articles studying medial thalamotomy using SRS for intractable pain. Only studies in which SRS was used to target the medial thalamus for pain were included. For centers with multiple publications, care was taken to avoid recounting individual patients. The literature review revealed 6 studies describing outcomes of medial thalamotomy using SRS for a total of 125 patients (118 included in the outcome analysis); 52 patients were treated for cancer pain across 3 studies, whereas 5 studies included 73 patients who were treated for non-malignant pain. The individual studies demonstrated initial meaningful pain reduction in 43.3 % to 100 % of patients, with an aggregate initial

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meaningful pain reduction in 65 patients (55 %) following SRS medial thalamotomy. This effect persisted in 45 patients (38 %) at the last follow-up; AEs were seen in 6 patients (5 %), which were related to radiation in 5 patients (4 %). The authors concluded that medial thalamotomy using SRS was effective for select patients with treatment-resistant pain and was remarkably safe when modern radiation delivery platforms were used; more posteriorly placed lesions within the medial thalamus were associated with better pain relief. These researchers stated that more studies are needed to shed light on differences in patient responses. Moreover, these researchers stated that future controlled studies in larger groups of patients are needed to draw definitive conclusions regarding the effectiveness of medial thalamotomy using SRS and to shed light on difference in patient's response that can be used to provide individualized treatment recommendations.

The authors stated that this review had several drawbacks. First, the small number of patients included, which made comparisons among different groups of patients difficult. Second, there was considerable variability in methods for reporting pain reduction across studies. In most studies, crude assessment methods, which have been criticized for their subjectivity, were employed and did not examine the affective dimension of pain. Third, there was also considerable heterogeneity in the operative techniques at many stages, including targeting delineation methods, reference imaging examination for targeting, device employed for SRS, dose delivered, number of isocenters, as well as method for lesion localization following SRS. Fourth, all studies were open-label, observational patient series; thus, subject to bias from the placebo effect. These researchers stated that controlled studies should be performed in patients with non-malignant pain syndromes.

Magnetic Resonance-Guided Focused Ultrasound Thalamotomy for the Treatment of Neuropathic Pain

Taranta et al (2023) stated that MRgFUS is an innovative therapeutic approach for medically refractory tremor. It is currently under investigation for other neurological diseases including refractory neuropathic pain (NP). In a systematic review, these investigators examined available evidence on the safety and effectiveness of MRgFUS in the treatment of refractory NP. Eligible studies were identified by searching published studies in PubMed and Scopus databases from inception to December 2022, and by identifying ongoing studies registered on the clinicaltrials.gov website. These researchers found 3 observational studies and 9 ongoing studies. In published studies, the involved population ranged from 8 to 46 patients with overall 66 patients being included with NP or trigeminal neuralgia. The target lesion was in the posterior part of the central lateral nucleus of the thalamus, bilaterally. Outcomes were assessed at different times via the VAS, showing a variable degree of improvement. AEs were rare, mild, and transient (vertigo, paresthesia, and dysesthesias) with intra-cerebral bleeding being reported as major AE in 1 case only. Among ongoing studies, these researchers found 3 prospective, randomized, sham-controlled, cross-over trials and 6 observational studies. Inclusion criteria are previous failure of more than 3 pharmacotherapies and NP duration of longer than 6 months. The thalamus is the main proposed target and measured outcomes are accuracy of the procedure and pain relief, with a follow-up period ranging from 1 week to 1 year. The authors concluded that MRgFUS may be a safe and effective alternative approach in the treatment of advanced NP. The findings of this systematic review showed that MRgFUS is a promising technique in the treatment of NP although there is a general lack of high-quality studies examining its safety and effectiveness. Preliminary data from observational studies on a limited number of subjects showed that MRgFUS, with the thalamus as the main target, was safe and effective for managing NP with minor side-effects; however, HIFU thalamotomy is not feasible for every patient or every case of NP. These researchers stated that much work remains to be carried out to determine in which cases MRgFUS would provide significant therapeutic and cost-effective value in the management of NP. Ongoing RCTs will provide more robust data to ascertain the benefits and risks of the procedure. These investigators stated that future studies should include larger and more homogeneous samples of patients in order to make outcomes comparable. Objective assessment tools to assess pain should be used together with effective follow-up strategies and schedules. Furthermore, concomitant drug abuse, that could influence the assessment, as well as the possible placebo effect experienced by patients, should be considered when evaluating the MRgFUS results on NP.

Ishaque et al (2023) noted that medial thalamotomy has been reported to benefit patients with neuropathic pain; however, widespread adoption of this procedure has been limited by reporting of clinical outcomes in

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studies without a control group. In a prospective, double-blinded RCT, these researchers aimed to minimize confounders associated with medial thalamotomy for the treatment of chronic pain by using modern MRI-guided stereotactic lesioning and a rigorous clinical design. This study included 10 patients with trigeminal neuropathic pain; and sham procedures were employed as controls. Subjects underwent assessments by a pain psychologist and pain management clinician, including use of the following measures: the Numeric Pain Rating Scale (NPRS); patient-reported outcome measures (PROMs); and patient's impression of improvement at baseline, 1 day, 1 week, 1 month, and 3 months post-procedure. Subjects in the treated group underwent bilateral FUS medial thalamotomy targeting the central lateral nucleus. Subjects in the control group underwent sham procedures with energy output disabled. The primary effectiveness outcome measure was between-group differences in pain intensity (using the NPRS) at baseline and at 3 months post-procedure; AEs were measured for safety and included MRI analysis. Exploratory measures of connectivity and metabolism were analyzed using diffusion tensor imaging, functional MRI, and PET, respectively. There were no serious complications from the FUS procedures. MRI confirmed bilateral medial thalamic ablations. There was no significant improvement in pain intensity from baseline to 3 months, either for patients undergoing FUS medial thalamotomy or for sham controls; and the between-group change in NPRS score as the primary effectiveness outcome measure was not significantly different. PROMs showed improvement (i.e., a decrease) only in pain interference with enjoyment of life at 3 months. There was a perception of benefit at 1 week, but only for patients treated with FUS and not for the sham cohort. Advanced neuroimaging revealed that these medial thalamic lesions altered structural connectivity with the post-central gyrus and showed a trend toward hypo-metabolism in the insula and amygdala. The authors concluded that this RCT of bilateral FUS medial thalamotomy did not lower the intensity of trigeminal neuropathic pain, although it should be noted that the ability to estimate the magnitude of treatment effects was limited by the small cohort.

Ahmed et al (2024) noted that MRgFUS central lateral thalamotomy (CLT) has not yet been validated for the treatment of refractory neuropathic pain (NP). In a prospective, non-randomized, single-arm, phase-I clinical trial, these researchers examined the safety and potential effectiveness of MRgFUS CLT for the treatment of refractory NP. This study included patients with NP for more than 6 months related to phantom limb pain, spinal cord injury (SCI), or radiculopathy/radicular injury and who had undergone at least 1 previous failed intervention. The main outcomes were safety profile and pain as assessed using the brief pain inventory (BPI), the pain disability index (PDI), and the numeric rating scale (NRS). Medication use and the functional connectivity of the default mode network (DMN) were also assessed. A total of 10 patients were enrolled, with 9 achieving successful ablation. There were no serious AEs, and 12 mild/moderate severity events. The mean age was 50.9 years (SD: 12.7), and the mean symptom duration was 12.3 years (SD: 9.7). Among 8 patients with a 1-year follow-up, the BPI decreased from 7.6 (SD: 1.1) to 3.8 (SD: 2.8), with a mean percent decrease of 46.3 (SD: 40.6) (paired t -test, $p = 0.017$). The mean PDI decreased from 43.0 (SD: 7.5) to 25.8 (SD: 16.8), with a mean percent decrease of 39.3 (SD: 41.6) ($p = 0.034$). NRS scores decreased from a mean of 7.2 (SD: 1.8) to 4.0 (SD: 2.8), with a mean percent decrease of 42.8 (SD: 37.8) ($p = 0.024$). Subjects with predominantly intermittent pain or with allodynia responded better than individuals with continuous pain or without allodynia, respectively. Some patients decreased medication use. Resting-state functional connectivity changes were noted, from disruption of the DMN at baseline to re-activation of connectivity between DMN nodes at 3 months. The authors concluded that MRgFUS CLT was feasible and safe for the treatment of refractory NP; and has potential utility in reducing symptoms as measured by validated pain scales.

Colasurdo et al (2024) stated that MRIgFUS represents a promising alternative for patients with chronic neuropathic pain who have failed medical management and other therapeutic options. Early single-center experience with chronic neuropathic pain and trigeminal neuralgia has demonstrated favorable long-term outcomes. Excellent safety profile with low risk of motor and sensory complications and so far anecdotal permanent neurologic deficits made FUS a powerful tool to treat patients who are otherwise hopeless. The authors concluded that neuromodulation may be the most influential factor driving outcomes and studies devised to detect neuroplasticity will be critical to guide such therapies. These investigators also provided an overview of ongoing clinical studies that will furnish further evidence in the near future.

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Focused Ultrasound Thalamotomy for the Treatment of Segmental Dystonia / Post-Operative Dystonia

Cummins et al (2024) MRIgFUS is an incisionless thermo-ablative procedure that may be used to treat medication-refractory movement disorders, with a growing number of potential anatomic targets and clinical applications. As of this article's publication, the only Food and Drug Administration (FDA)-approved indications of FUS for movement disorders are thalamotomy for ET and tremor-dominant PD, as well as pallidotomy for other cardinal symptoms of PD. These investigators presented a state-of-the-art review on all non-FDA approved indications of FUS for movement disorders, beyond the most well-described indications of ET and PD. The objective was to summarize the safety and effectiveness of FUS in this setting and provide a road map for future directions of FUS for movement disorders. A total of 25 studies on 172 patients were included. In patients with tremor plus dystonia syndromes (n = 6), VIM-FUS gave greater than 50 % tremor reduction, with no improvement in dystonia and worsened dystonia in 2/6 patients. Ventral-oralis complex (VO)-FUS gave greater than 50 % improvement for focal hand dystonia (n = 6), and 100 % return to musical performance in musician's dystonia (n = 6). In patients with MS and tremor (n = 3), improvement in tremor was observed in 2 patients with a favorable skull density ratio; no MS disease change was noted after VIM-FUS. In patients with tremor and co-morbid ataxia syndromes (n = 3), none was found to have worsened ataxia following VIM-FUS; all had clinically significant tremor improvement. STN-FUS for PD (n = 49) gave about 50 % improvement in PD motor symptoms, with dystonia and mild dyskinesias as possible adverse effects. CTT-FUS for ET (n = 42) gave 55 % to 90 % tremor improvement, with gait dysfunction as a rare persistent adverse effect. Pallido-thalamic tract (PTT-FUS) for PD (n = 50) gave approximately 50 % improvement in motor symptoms, with mild speech dysfunction as a possible adverse effect. The authors concluded that VIM-FUS appeared safe and effective for heterogenous tremor etiologies, and VO-FUS appeared most effective for isolated segmental dystonia. STN-FUS was effective for PD symptom reduction; post-operative dystonia and mild on-medication dyskinesias required medical management. Tractography-based targeting with CTT-FUS for ET and PTT-FUS for PD showed promising early results. Moreover, these researchers stated that larger prospective studies with long-term follow-up are needed to the examine the safety and effectiveness non-FDA approved indications for FUS.

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